From single mutants to a calibrated landscape

Better fitness distributions. Similar search rankings.

Singles and a FASTA file are insufficient. This model also needs measured combinations from the same assay.

1 Define a small mutation library

The FASTA supplies residue identities and position numbers. The assay supplies wild-type fitness and single-mutant fitness. Use one-based positions.

Choose the positions and allowed residues first. Each allowed substitution needs a measured single. A full-protein DMS usually defines too many combinations.

2 Measure random combinations

Sample uniformly from combinations with at least two substitutions. Measure them on the same fitness scale as the wild type and singles.

Our stronger evidence uses 192 combinations. A pilot selected for high fitness would bias the fitted distribution. These measurements are an extra experimental cost.

3 Fit the existing pilot model

Singles set the individual mutation effects. The pilot fits two terms for mutation count and calibrates the strength of random pair effects.

A sigmoid maps latent scores to fitness. Each seed supplies fixed pair effects. The fit does not learn which specific pairs interact.

4 Match the pilot distribution

Each draw scores every supported combination. The new model maps each score's percentile to the same percentile of measured pilot fitness.

This changes fitness spacing. Combination order stays the same, except for added ties. Wild type and singles retain their measured values exactly.

Generated score → **Percentile** → **Pilot fitness quantile**

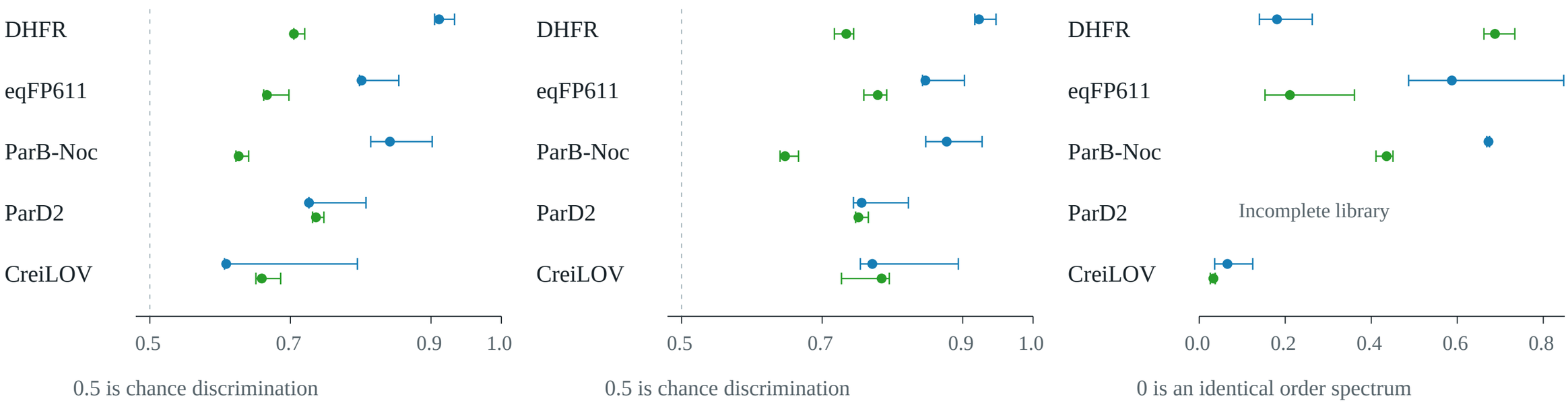
```
f_new(x) = Q_pilot( F_draw( f_pilot(x) ) )
```

Example: 80th percentile maps to 80th percentile.

What changed across five proteins?

Both models use the same 192 calibration combinations. Classifier tests use 1,000 separate genotypes per protein.

– Current pilot – Quantile-calibrated pilot Points show medians. Bars show the range across three draws.



C2ST improves on three proteins. Values near 0.7 still show detectable mismatch. AUC is not classification accuracy. There is no universal passing value.

Spectral distance improves on three of four proteins. DHFR worsens from 0.181 to 0.688. A nonlinear fitness map can change interaction-order variance.

Search rankings change little

We tested 12 methods, ten search seeds, and budgets of 48 or 96 queries. Both models use 192 extra calibration measurements.

Measure	Current pilot	New model
Agreement on clear method winners	93.7%	94.3%
Kendall rank correlation	0.628	0.653

Only method pairs with a clear measured difference count towards winner agreement. Means give each protein and query budget equal weight.

Use the model

```
from mutascape.experimental import (
    QuantileCalibratedLandscape,
)
model = QuantileCalibratedLandscape.fit(
    sequence, singles, pilot)
landscape = model.draw(seed=42)
fitness = landscape.evaluate_batch(variants)
```

Both tables need mutations and fitness columns. Include wild type as {} in singles. Repeated queries return the same fitness.

Limits. One calibration set per protein. Draw ranges are not confidence intervals. Spectra include calibration genotypes. CreiLOV uses a binary projection. Exact enumeration defaults to one million variants.

Sources. Lopez-Paz and Oquab, ICLR 2017, arxiv.org/abs/1610.06545. Model and data in mutascape.experimental.quantile and results/quantile_confirmation_20260923. These results compare two pilot models, not the earlier singles-only benchmark.